



Thin Layer Chromatography

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Abstract

In many experiments, it is important to be able to separate a mixture into its chemical components in order to isolate one compound or to assess the purity of the mixture. Thin layer chromatography (TLC) is one of the easiest and most versatile methods of doing this because of its low cost, simplicity, quick development time, high sensitivity, and good reproducibility. TLC is used by many industries and fields of research, including pharmaceutical production, clinical analysis, industrial chemistry, environmental toxicology, food chemistry, water, inorganic, and pesticide analysis, dye purity, cosmetics,

plant materials, and herbal analysis. In its simplest form, glass plates are coated with a uniform layer of silica gel (SiO_2). The dissolved sample is placed on the plate, and the plate is inserted into a screw-top jar containing the developing solvent and a piece of filter paper. When the solvent has risen to near the top of the plate, the plate is removed, dried, and visualized using UV light. Variations on this protocol are used for different purposes, including pretreating the sample, changing the sorbent, plate material, the solvent system, the development techniques, and method of detection and visualization or by coupling TLC to other techniques.



1. THEORY

TLC is a type of liquid chromatography in which the mobile phase is liquid and the stationary phase is a thin layer of material on top of a flat plate. This layer of material is known as the sorbent. The mobile phase is known as the developing solvent, and it transports the solutes through the stationary phase. The speed at which the solute moves through the stationary phase depends on the force of the mobile phase as it dissolves the solute and moves it up the plate, and the resistance of the sorbent as it pulls the solute out of solution and back into the sorbent. Molecules move along the plate with a stop-and-go motion as the solute becomes repeatedly absorbed and desorbed. Because of this, only a small fraction of the solute is moving at any time, but each spot will travel a mean distance. The separate zones will spread out because of the random fluctuations and movements of individual particles and variations in the uniformity of the sorbent. Substances that are strongly attracted to the sorbent move more slowly because they spend more time in the sorbent, and those that move more quickly are less attracted to the solid layer, are more soluble in the mobile phase, and spend more time there (Fried and Sherma, 1999). Therefore, compounds with different properties can be separated from one another by exploiting the diverse interactions of the solutes with the sorbent and the mobile phase.

There are many ways in which TLC can be classified. It can be classified by the mechanism of separation: adsorption (physical sorption of the solutes onto the sorbent particles), partition (dissolution of the solutes into a stationary liquid on the sorbent), ion exchange (attraction of ions to groups of opposite charge on the sorbent), and size exclusion or gel permeation (retention or rejection based on size or shape). Most often, TLC involves adsorption and partition because they both involve the same type of forces, induced dipole–dipole and hydrogen bonding. In addition, actual separations can involve all of these mechanisms. However, adsorption TLC is most sensitive

to the differences in configuration that affect the free energy of absorption making it especially useful for separating compounds on the basis of differences in polarity. In contrast, partition-based separations differentiate compounds on the basis of solubility ([Fried and Sherma, 1999](#)).

In normal- or direct-phase TLC, the sorbent is polar, so the more polar solutes move more slowly and stay closer to the origin, while the nonpolar solutes will move more quickly and closer to the solvent front. By increasing the polarity of the mobile phase, the polar solutes can be drawn farther from the origin to increase separation. The opposite is true for reverse-phase TLC (RP-TLC). A nonpolar sorbent will bind nonpolar solutes better and they will migrate up the plate slowly, and the polar components will move quickly near the solvent front. To increase the distance traveled by the nonpolar zones, the polarity of the developing solvent must be decreased.

Paper chromatography (PC), gas chromatography (GC), and high-performance liquid chromatography (HPLC, see [Reverse-phase HPLC Analysis and Purification of Small Molecules](#)) are three other methods commonly used to separate mixtures. Though PC and TLC are very similar (both are open systems where the mobile phase moves over the sample to separate it into its constituents which are detected after development of the system), TLC is easier to resolve, faster, and more sensitive. Consequently, as TLC has become more common, PC is fading from use.

Gas chromatography is difficult to compare with TLC because, unlike HPLC, TLC, and PC, the mobile phase of GC is a gas, the stationary phase is a liquid or polymer column, the concentration of the compound in the gas phase is only a function of the vapor pressure of the gas, very high temperatures are used, and the components are separated mostly on the basis of boiling point. Therefore, TLC can be most easily compared with HPLC.

There are some ways in which HPLC is superior to TLC. A theoretical plate is a hypothetical zone found in a separation process in which two phases, such as liquid and vapor, have established a state of equilibrium with one another. The more theoretical plates, the more efficient the separation. A typical HPLC column will give up to 10 000–20 000 theoretical plates while TLC can provide only 5000 theoretical plates in a typical 3–6 cm development distance ([Fried and Sherma, 1999](#)). For this reason, TLC results in a less efficient separation than HPLC. Also, unless forced-flow TLC is used to control the mobile phase velocity, the speed at which the mobile phase moves through the plate may not be the same every time. Because HPLC can control the mobile phase velocity, it is more precise. Furthermore, because HPLC solvent systems are well-equilibrated gradients, the solvents will not separate during the experiment as can occur in TLC.

Though each round of TLC is more labor-intensive than a round of HPLC, more samples can be analyzed simultaneously using TLC, allowing it to be a more high-throughput technique than HPLC, as well as decreasing the cost per analysis. Simultaneous analysis of samples and standards also statistically increases analytical precision and accuracy. On a TLC plate, the problem of disappearing or unexpected peaks is nonexistent because the entire sample is applied and can be detected visually. Moreover, using HPLC and GC, the highly retained substances form the widest peaks and have the worst resolution, but with TLC, the substances that are retained the longest (lowest R_f) form the tightest zones and are detected with the highest sensitivity. Even though the maximum number of substances that can be completely separated is higher for HPLC than classical TLC (20–40 for HPLC versus 10–20 at most for TLC), using two-dimensional TLC, up to 100–250 can be completely resolved. Under forced-flow conditions, this number can be increased to over 500 (Fried and Sherma, 1999).

TLC is also more versatile than HPLC. Many more solvent systems can be used for the developing solvent because they are totally evaporated before the solute is detected, and the choice of solvent for the sample is less important because it is removed before the mobile phase is added. In contrast, the solvent used to dissolve the sample in HPLC must be compatible with the mobile phase. TLC plates can be impregnated with various reagents, and the many methods of development (continuous, multiple, circular, and anticircular) as well as the many available solvents allow the system to be optimized for the separation of only the compounds of interest. HPLC has fewer available stationary phases than TLC, and a more limited choice of developing methods and solvents. Though detection limits are about the same for HPLC and TLC (picogram to microgram) (Fried and Sherma, 1999), detecting the samples is much easier for TLC, and there are many more methods for doing it with TLC compared to HPLC. After the completion of the experiment, TLC requires less clean up than HPLC because, unlike HPLC columns, the TLC plates are not reused. There is less liquid waste as well because solvent use for TLC is much less than for HPLC. Although there are reasons to choose HPLC over TLC, TLC provides a versatile, cheap, and simple method for analytical and preparative applications.



2. EQUIPMENT

Glass screw-top jar or developing tank
TLC plate

Glass cutter
Drying oven, dessicator cabinet, or drying rack
UV light
Sprayer
Spray chamber
Microcapillary tube
Tweezers
Scissors
Ruler
Pencil



3. MATERIALS

Possible developing solvents for mobile phase solvent system:

Benzene
Toluene
Hexane
Methanol
Acetic acid
Ammonium hydroxide
Acetone
Chloroform
Pentane
Isooctane
Acetonitrile
Methyl-*t*-butyl ether
1,2-Dichloroethane
Methylene chloride
Diethyl ether
Tetrahydrofuran
Water
Isopropanol
Ethanol
Ethyl acetate
Dioxane

Visualization reagents:

Iodine crystals
7,8-Benzoflavone

Ethanol
 Sulfuric acid
 Potassium dichromate
 Nitric acid
 Cupric acetate
 Phosphoric acid
 Cupric sulfate
 Water
 Ammonium sulfate
 Phosphomolybdic acid
 Rhodamine B
 Antimony trichloride
 Antimony pentachloride
 Vanillin
 Potassium hydroxide
 Fluorescein sodium
 2',7'-Dichlorofluorescein
 pH indicators:
 Bromocresol green
 Bromothymol blue
 Bromophenol blue
 Methyl red
 Malachite green

3.1. Solutions & buffers

Step 4 Solution for Decreasing the Fading Time of Iodine

Component	Final concentration	Stock	Amount
7,8-benzoflavone	3 mg ml ⁻¹		0.3 g
Ethanol	95%		95 ml
Sulfuric Acid	1.5%	30%	5 ml

Potassium Dichromate Charring Solution

Component	Final concentration	Stock	Amount
Potassium Dichromate	50 mg ml ⁻¹		5 g
Sulfuric Acid	40%	40%	100 ml

Cupric Acetate Charring Solution

Component	Final concentration	Stock	Amount
Cupric Acetate	0.03%		3 g
Phosphoric Acid	8%	8%	100 ml

Cupric Sulfate Charring Solution

Component	Final concentration	Amount
Cupric Sulfate	0.1 g ml ⁻¹	100 g
Phosphoric Acid	0.8%	80 ml

Add to 800 ml water and then bring the volume to 1 l. Stir on a magnetic stir plate for 1 h

Phosphomolybdic Acid Solution

Component	Final concentration	Amount
Phosphomolybdic Acid	5%	5 g
Ethanol		100 ml

Rhodamine B Solution

Component	Final concentration	Amount
Rhodamine B	0.5 mg ml ⁻¹	50 mg
Ethanol		100 ml

Rhodamine 6G Solution

Component	Final concentration	Stock	Amount
Rhodamine 6G	0.5 mg ml ⁻¹		50 mg
Ethanol	96%	96%	100 ml

Antimony Chloride Solutions

Component	Final concentration	Amount
Antimony trichloride or pentachloride	10–20%	10–20 g
Chloroform or Carbon Tetrachloride		100 ml

Anisaldehyde-Sulfuric Acid Solution

Component	Final concentration	Amount
Anisaldehyde	1%	1 ml
Sulfuric acid	2%	2 ml
Glacial Acetic Acid	98%	100 ml

Anisaldehyde-Sulfuric Acid Solution for Spraying

Component	Final concentration	Amount
Anisaldehyde	0.5%	0.5 ml
Sulfuric acid	7.7%	8 ml
Methanol	82%	85 ml
Glacial Acetic Acid	9.6%	10 ml

Add anisaldehyde and sulfuric acid to methanol and acetic acid while cooling in an ice bath

Vanillin-Phosphoric Acid Solution

Component	Final concentration	Amount
Vanillin	1.2%	1 g
Ethanol	29%	25 ml
Water	29%	25 ml
o-Phosphoric Acid	41%	35 ml

Vanillin-Isopropanol Solution

Component	Final concentration	Amount
Vanillin	2%	2 g
Isopropanol		100 ml

Ethanol KOH Solution (for dipping plates after spraying with Vanillin-Isopropanol Solution)

Component	Final concentration	Stock	Amount
KOH	10 mM	1 M	1 ml
Ethanol			100 ml

Fluorescein Solutions

Component	Final concentration	Amount
Fluorescein Sodium or 2',7'-Dichlorofluorescein	0.05–0.2%	5–20 g
Ethanol		100 ml

pH Indicator Solutions

Component	Final concentration	Amount
pH Indicator	0.01–1%	1–10 g
Water or Water and Ethanol		100 ml

Caution Take appropriate measures when working with strong acids. Work in a fume hood and wear a lab coat and gloves. Clean up spills immediately.



4. PROTOCOL

4.1. Preparation

Prepare the necessary solutions.

4.1.1 Choose a TLC sorbent

The most common TLC sorbents are silica-based. Silica gel (SiO_2) is a white porous material made by precipitation from silicate solutions by addition of acid (Wall, 2005). It consists of bonded silicon and oxygen with residual hydroxyl groups. Many sorbents with standardized particle sizes can be obtained commercially bound to plastic, aluminum, or glass plates. Silica gel plates can separate all classes of compounds and can be impregnated with other compounds to increase selectivity. Non-silica-gel-based sorbents include: kieselguhr (composed of the remains of diatoms whose cell wall is composed of silica), cellulose (glucopyranose units joined together by oxygen bridges), polyamide (produced from polycaprolactam, polyhexamethyldiaminoadipate, or polyaminoundecanoic acid), and aluminum oxide (Al_2O_3 ; also called alumina, it can be manufactured in three different pH ranges). The table below describes the different types of sorbents and the classes of compounds they can separate (Wall, 2005).

Sorbent	Modification	Classes of compounds able to separate
Silica gel		All classes of compounds
	Amino-bonded	Carbohydrates, aflatoxins, herbicides, and tetracyclines
	Cyano-bonded	Many classes. Particularly, pesticides, steroids, and preservatives
	Diol-bonded	Many classes. Particularly, steroids and hormones
	Reversed-phase	Steroids, tetracyclines, phthalates, antioxidants, lipids, barbiturates, capsaicins, aminophenols, and fatty acids
	Chiral modified	Enantiomers of amino acids, halogenated, <i>N</i> -alkyl, and α -methyl amino acids, simple peptides, and catecholamines
	Impregnated with silver nitrate	Lipids
	Impregnated with caffeine	Polyaromatic hydrocarbons
	Impregnated with boric acid/phosphate	Carbohydrates
Kieselguhr		Carbohydrates, aflatoxins, herbicides, and tetracyclines
Cellulose		Amino acids and derivatives, food dyes, and carbohydrates
Polyamide		Phenols, flavonoids, and nitro-compounds
Aluminum oxide		Basic compounds, steroids, terpenes, and aromatic and aliphatic hydrocarbons

4.1.2 Choose a developing solvent system

The mobile phase used for silica gel and modified silica gel plates is usually one solvent or one solvent with a polar solvent added to increase the polarity of the mobile phase. To design a good solvent system, the sample should be spotted on the plate of choice, and tested in beakers or small jars with 1 cm of only one solvent in the bottom. The solvent strength should be increased using the eluotropic series (below) until a solvent brings the sample about halfway up the plate. The eluotropic series was designed by Lloyd Snyder who defined solvent strength as depending on the physical characteristics

of the liquid as well as the sorbent being used. This value ϵ^0 , is defined as the adsorption energy per unit of standard solvent for the given solvent/sorbent combination (Snyder, 1964; 1966). To improve resolution, another solvent of a similar strength should be added, but it should be from a different selectivity group (developed by Snyder and Glajch). The most improvement is usually seen when the added solvent is from a very different selectivity group. Solvent systems with the same strength but made up of different solvents will give very different resolutions and selectivities (Snyder and Glajch, 1981a; b; 1982).

4.1.3 Eluotropic series

Solvent	Calculated solvent strength on silica gel (ϵ^0)
<i>n</i> -Pentane	0.00
<i>n</i> -Hexane	0.01
Cyclohexane	0.03
Cyclopentane	0.04
Carbon tetrachloride	0.14
Toluene	0.22
Benzene	0.25
Chloroform	0.31
Dichloromethane	0.32
Tetrahydrofuran	0.35
Diethyl ether	0.38
Acetone	0.43
Ethyl acetate	0.45
Aniline	0.48
Acetonitrile	0.50
Pyridine	0.55
Ethanol	0.68
Methanol	0.73
Acetic acid	<1
Water	>>1

4.1.4 Solvent selectivity groups according to Snyder and Glajch (ref)

Group	Solvents
I	Aliphatic ethers
II	Aliphatic alcohols
III	Pyridine, tetrahydrofuran, glycol ethers, and amides (except formamide)
IV	Formamide, acetic acid, and glycols
V	Dichloromethane and 1,2-dichloroethane
VI	Aliphatic ketones and esters and 1,4-dioxanacetonitrile
VII	Aromatic hydrocarbons and ethers and aromatic halo- and nitro-compounds
VIII	Chloroform, water, nitromethane, and m-cresol

4.1.5 Choose a method of detection and visualization

A good time to try out methods of detection and visualization is immediately after choosing a developing solvent system. The methods can be tested using the plates that were used for selecting the solvent system. It is important that the solvents be totally evaporated before detection and visualization. The plates can be air-dried in a well-ventilated area placed on a table or a drying rack. They can also be dried with a hair dryer or in an oven. When drying at high temperatures, it is important to make sure that volatile sample compounds do not evaporate and that the temperature is not so high that the sample decomposes.

There are many methods for detection and visualization. Colored compounds can be seen on the plate in visible light. Fluorescent compounds can be detected using long wave UV (366 nm) light. Compounds on the plate can also be derivatized by fluorogenic reagents (Kovar and Morlock, 1996). When an inorganic phosphor is incorporated into the sorbent and the plate is viewed at short wave (254 nm) UV light, the nonfluorescent developed sample zones should show up as darker spots or zones on the plate. Detection with UV light should be done in a dark area or in a viewing cabinet constructed from a cardboard box.

Another method of detection involves spraying the plate (using a spray bottle or atomizer) with a reagent that allows visualization of the developed zones on the plate. There are many reagents that can be used for spraying, and many can be obtained commercially. Sometimes, dipping the plates in

the reagent is more effective and is safer than spraying. Though some of these reagents reversibly mark the plates, others are nonreversible, and some charring reagents should be used only on glass-backed plates because they are corrosive and will warp and/or destroy plastic and metal-backed plates. The detection limits for TLC are in the range of 0.1–1 μg . The following table describes some detection reagents and their properties (Fried and Sherma, 1999).

Reagent	Selective for	Method of detection	Other
Iodine	Universal	Closed container containing plate and a few iodine crystals	The spots that result from detecting with Iodine crystals fade quickly. The rate of fading can be decreased using the Solution for Decreasing the Fading Time of Iodine (Fried and Sherma, 1999)
Potassium dichromate charring solution	Universal	Spray plate and heat to 120–280 °C for 15–40 min	Fried and Sherma (1999)
Sulfuric acid	Universal	Spray plate and heat to 120–280 °C for 15–40 min	Also, 50% aqueous sulfuric acid, 1:1 nitric acid: sulfuric acid, 3:1 acetic anhydride: sulfuric acid (Fried and Sherma, 1999)
Cupric acetate charring solution	Unsaturated phospholipids (Sherma and Bennett, 1983)	130–180 °C for up to 30 min	Suitable for use with some commercial organic bound layers (Fried and Sherma, 1999)
Curpic sulfate charring solution	Saturated and unsaturated phospholipids (Sherma and Bennett, 1983)	Dipped and heated for 10–15 min at 160 °C	Fried and Sherma (1999)

Continued

Phosphomolybdic acid solution	Steroids, bile acids, lipids, fatty acids, substituted phenols, indole derivatives, prostaglandins, essential oil components, and drugs (Jork et al., 1993)	Heat 5–20 min at 110 °C	Exposure of plate to ammonia vapors decolorizes the background of the plate improving contrast (Fried and Sherma, 1999)
Rhodamine solution	Rhodamine B: many organics including lipids Rhodamine 6G: lipophilic substances	Spray	For Rhodamine 6G, may be fluorescent under long wave light. It can also be incorporated in the sorbent or in the mobile phase (Fried and Sherma, 1999)
Antimony trichloride or pentachloride solution	Vitamins, water- and fat-soluble pigments, sterols, terpenes, steroids, bile acids, sapogenins, glycosides, and phospholipids (Jork et al., 1990)	Spray and heat at 110 °C	Spots fluoresce in long wave UV light (Fried and Sherma, 1999)
Anisaldehyde-sulfuric acid solution	Antioxidants, steroids, prostaglandins, carbohydrates, phenols, glycosides, sapogenins, essential oil components or terpenes, antibiotics, and mycotoxins	Dip and heat at 90–125 °C for 1–15 min	Universal reagent for natural products and different colors produced can help in identifying the compound (Jork et al., 1990; Fried and Sherma, 1999)
Vanillin-phosphoric acid and vanillin-potassium hydroxide	Steroids, sterols, triterpenes, cucurbitacins, digitalis glycosides, prostaglandins, and saponins	Dipped or sprayed and heated to 120–160 °C for 5–15 min	Colored zones on a pale background that often will fluoresce under long wave UV light (Jork et al., 1990; Fried and Sherma, 1999)

Continued

Fluorescein sodium and 2',7'-dichlorofluorescein	Saturated and unsaturated organic lipophilic compounds	Spray	Fluorescent spots on a purple background in UV light (Fried and Sherma, 1999)
pH indicators	Acidic or basic compounds in the layer	Spray	Fried and Sherma (1999)

4.1.6 Choose a development chamber

There are many types of TLC development chambers that have been designed with features to help control development reproducibility. Of course, the chamber should be big enough to fit the plate. For small analytical TLC plates, any small glass screw-top jar will suffice. Chambers without screw tops should be closed using vacuum grease to keep the chamber as sealed as possible.

4.1.7 Choose a method of development

Normal-phase and reverse-phase developments are simple. After placing the plate in the developing chamber and closing the lid, the plate is developed until the solvent reaches near the top of the plate. Then it can be removed for visualization.

Besides normal-phase and reverse-phase developments, there are many ways of developing the sample. In single mobile phase multiple development, the mobile phase is allowed to travel up the plate to a predetermined point. Then the plate is removed, dried, put back into the development chamber, and allowed to develop up to the same point. This is repeated until the sample components have migrated enough to be resolved. This is often used in amino acid analysis because there are so many components in the sample that the components have not migrated far enough to be resolved after a single development ([Wall, 2005](#)).

When the plate is not long enough for sufficient separation, continuous development can be used. During continuous development, only a single solvent is used in the mobile phase. The mobile phase travels up the plate, and at the top of the plate, provisions are made for the solvent to evaporate at the lid of the chromatography chamber. As the solvent evaporates from the top of the sorbent layer, the mobile phase continuously flows through the plate. This is a simpler process than single phase mobile phase development, but the downside is that it can decrease the resolution ([Wall, 2005](#)).

Two-dimensional development is a powerful separation technique traditionally used to separate amino acids (Meloan, 1999). The sample is spotted on the plate alongside some standards and run normally in the first dimension. The plate is turned 90° and run in another solvent system. To obtain the best separation, the two solvent systems should be very different. The solvent system for the first dimension should separate at least half of the components and it should be more polar than the second solvent system. The second solvent system should provide similar resolution to the first while separating the remainder of the components of the sample. The final plate will be a fingerprint and the spots can be compared with standards for identification (Wall, 2005).

When the sample has both very polar and very nonpolar components, manual gradient development can be used. The first solvent used is the most polar. It is allowed to travel slightly up the plate. Then, the plate is dried and the solvent is replaced with one slightly less polar which travels slightly farther up the plate. This is repeated as many times as necessary using a less polar solvent every time until separation is achieved. When this process is automated, it is known as automated multiple development (Wall, 2005).

4.1.8 TLC coupling techniques

It will not be discussed in depth here, but TLC can be coupled to other methods of separation or identification. These couplings include TLC and HPLC, TLC and MS, TLC and FTIR (both on-line and off-line), and TLC and Raman spectroscopy.

4.2. Duration

Preparation	1 h
Protocol	2–4 h

See Fig. 24.1 for the flowchart of the complete protocol.



5. STEP 1 PREPARING THE TLC PLATES

5.1. Overview

Cut the TLC plate to an appropriate size and clean it in order to improve the quality of the separation.

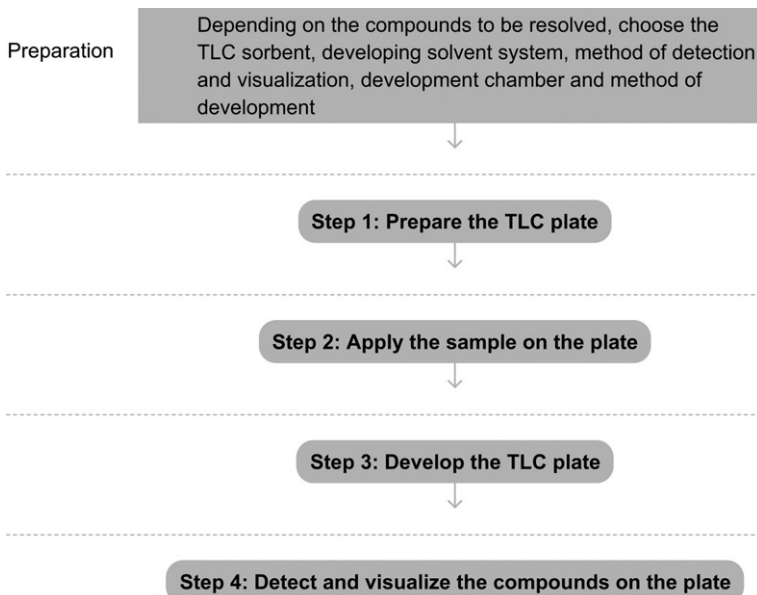


Figure 24.1 Flowchart of the complete protocol, including preparation.

5.2. Duration

1 h

- 1.1 Decide the dimensions of the TLC plate.
- 1.2 Cut the plates to the decided dimensions. Glass plates can be cut with a glass cutter.

Aluminum and plastic-backed plates should be cut with scissors. See [Video 1. <http://dx.doi.org/10.1016/B978-0-12-420067-8.00024-6>](http://dx.doi.org/10.1016/B978-0-12-420067-8.00024-6).
- 1.3 If the relative humidity is greater than 60%, heat the plate in a drying oven to dry off the excess moisture.
- 1.4 Place the TLC plate in the developing chamber containing ~1 cm of the mobile phase in the bottom. Allow the solvent to reach the top of the plate to remove impurities from the plate and to increase resolution.

The plate is now ready for use.

See [Fig. 24.2](#) for the flowchart of Step 1.

6. STEP 2 SAMPLE APPLICATION

6.1. Overview

Dissolve sample in a solvent and apply it to the plate.

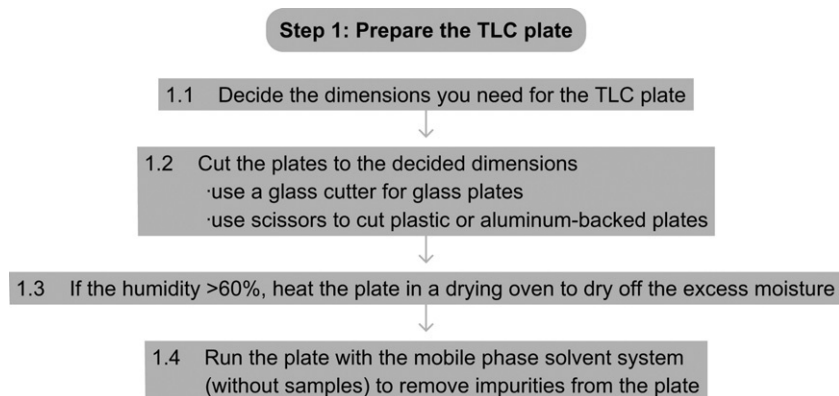


Figure 24.2 Flowchart of Step 1.

6.2. Duration

15 min

- 2.1 Dissolve the sample in the least polar single solvent, or mixture of solvents, in which all of the components are completely soluble. This will help keep the spot size small.
- 2.2 Make a mark about 1 cm from the bottom of the plate with the pencil.
- 2.3 Using a microcapillary tube, apply 1–5 μl of the dissolved sample onto the prepared TLC plate on the line just drawn. See [Video 2. <http://dx.doi.org/10.1016/B978-0-12-420067-8.00024-6>](http://dx.doi.org/10.1016/B978-0-12-420067-8.00024-6).
- 2.4 Let the spot dry.
- 2.5 Repeat this process until about 0.1 μg per spot, or 2–10 μg per 10 mm band, has been added. It is important to not add too much sample because overspotting can decrease resolution.
- 2.6 Make sure that the sample spot is completely dry before continuing. See [Fig. 24.3](#) for the flowchart of Step 2.



7. STEP 3 DEVELOPMENT OF THE PLATE

7.1. Overview

Develop the TLC plate. See [Video 3. <http://dx.doi.org/10.1016/B978-0-12-420067-8.00024-6>](http://dx.doi.org/10.1016/B978-0-12-420067-8.00024-6).

7.2. Duration

5 min to a few hours (depending on plate size)

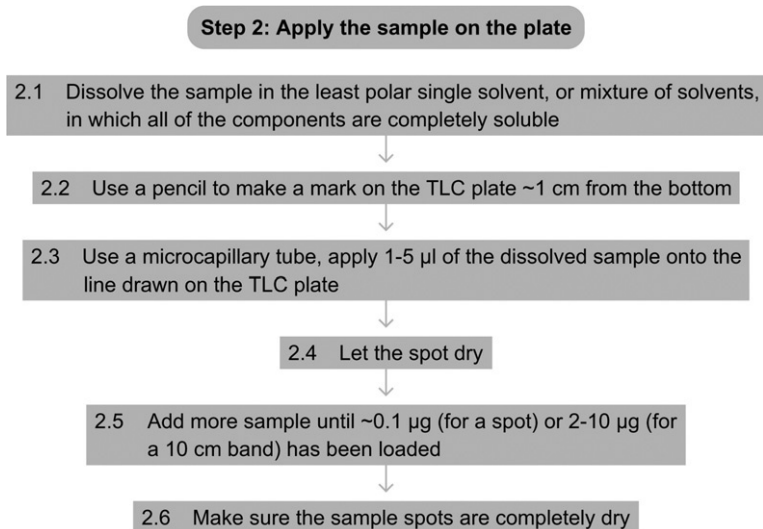


Figure 24.3 Flowchart of Step 2.

- 3.1 Make sure that the developing chamber is clean. Rinse it with acetone and let it dry.
- 3.2 Pour about 1 cm of mobile phase solvent system in the bottom of the developing chamber.
- 3.3 Place a piece of filter paper or Whatman paper inside the chamber. It should not be touching the solvent in the bottom of the jar or the plate.
- 3.4 Add the plate with the samples spotted on it, placing it in the chamber as close to vertical as possible.
- 3.5 Close the chamber tightly.
- 3.6 Wait until the solvent reaches a predetermined level (for small plates, usually about 1 cm from the top of the plate). Depending on which method of development was chosen, there may be more steps to the development process.
- 3.7 Remove the plate from the chamber.

7.3. Tip

It is best to find a way to curve the paper around the inside walls of the chamber so that it can hold itself up inside the jar. The filter paper keeps the air around the plate saturated with solvent. Make sure that the plate is still visible.

See [Fig. 24.4](#) for the flowchart of Step 3.

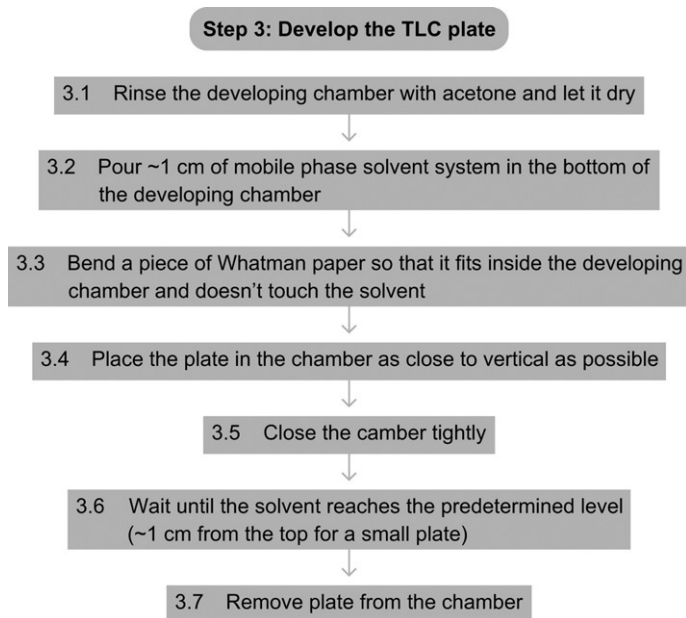


Figure 24.4 Flowchart of Step 3.



8. STEP 4 DETECTION AND VISUALIZATION OF THE SAMPLE

8.1. Overview

Using the method chosen previously, detect and visualize the developed TLC plate. See [Video 4. <http://dx.doi.org/10.1016/B978-0-12-420067-8.00024-6>](http://dx.doi.org/10.1016/B978-0-12-420067-8.00024-6).

8.2. Duration

Instant to 30 min, depending on the detection method

- 4.1** Make sure that plate is completely dry before proceeding.
- 4.2** If detecting with UV light, results are immediate, repeatable, and plates can be stored for an indefinite time. If detecting with another reagent, spray the plate or dip in the required reagent.
- 4.3** Once the spots are visible, outline the zones with a pencil because the spots often fade.
- 4.4** Photograph the TLC plate and store it or throw it away.

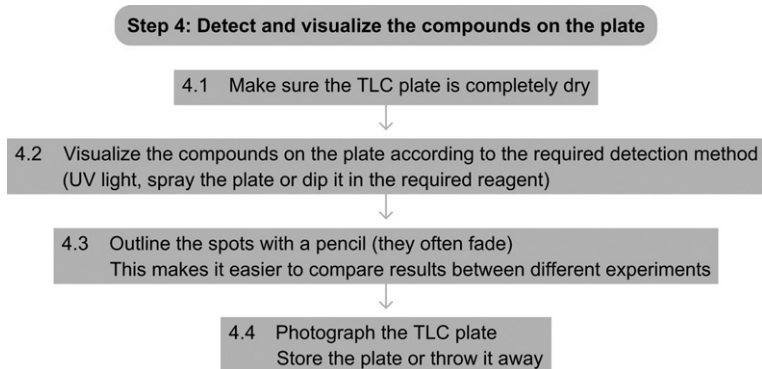


Figure 24.5 Flowchart of Step 4.

8.3. Tip

Follow the directions outlined in the table above for visualization using the appropriate reagent, heating at the correct temperature, if necessary. If spraying, make sure to spray past the edge of the plate before spraying again to ensure that an even coat is applied.

8.4. Tip

By marking the spots with a pencil, it is easier to compare the results with past and future plates.

8.5. Tip

The plate can also be photographed before you outline the spots with a pencil.

See Fig. 24.5 for the flowchart of Step 4.

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